

# *Week 10*

## Keel Bone Chicken Project

08/10/26



College of Engineering

# ***This Week***

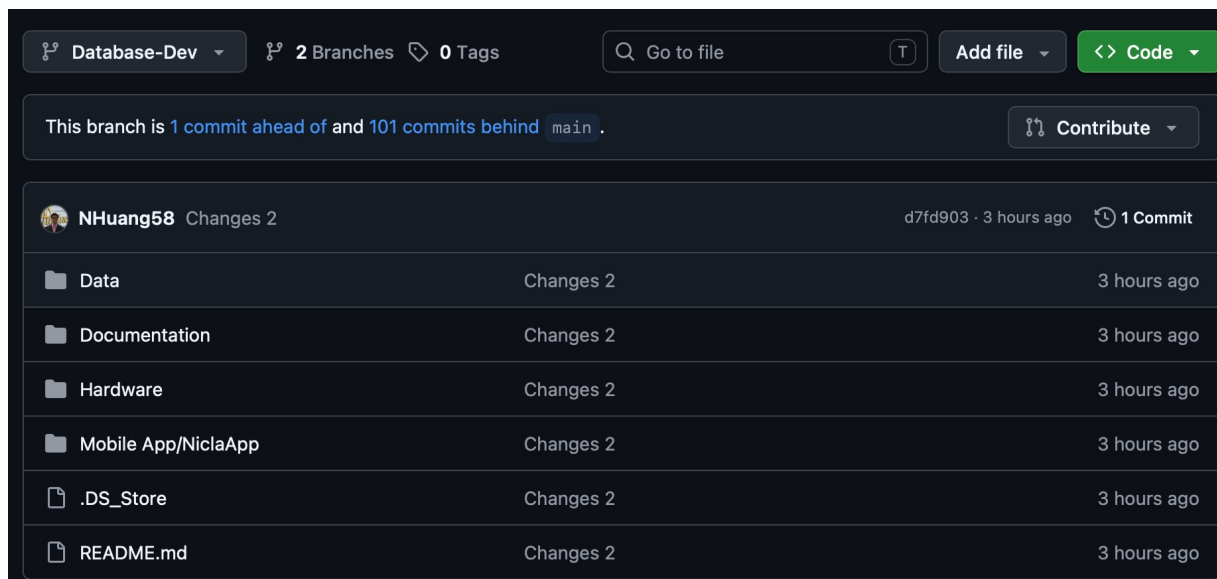
## Weekly Progress

- **Web Logger Updates**
  - Logger complete
- **PCB Updates**
  - Working on placement file updates
- **Documentation Updates**
  - GitHub Documentation Created
  - GitHub Files Transferred
  - Ownership Transferred
- **Paper Updates**
  - Paper Acceptance Date Changed
- **Next Week**
  - Plans for next week
  - Overall Thoughts

# Web Logger Updates

## Logger Complete

- All features in Web Logger
- Logger in separate branch of GitHub
- Changed database to SQLite3 database
  - Everything functional



The screenshot shows a GitHub repository interface for a branch named 'Database-Dev'. At the top, it indicates '2 Branches' and '0 Tags'. A search bar is present with the text 'Go to file'. To the right are buttons for 'Add file' and '<> Code'. Below this, a status bar shows 'This branch is 1 commit ahead of and 101 commits behind main'. A 'Contribute' button is also visible. The main content area shows a commit by user 'NHuang58' with the message 'Changes 2', committed '3 hours ago' with hash 'd7fd903'. Below the commit message is a table of files that were changed in this commit.

| File                | Changes   | Time        |
|---------------------|-----------|-------------|
| Data                | Changes 2 | 3 hours ago |
| Documentation       | Changes 2 | 3 hours ago |
| Hardware            | Changes 2 | 3 hours ago |
| Mobile App/NiclaApp | Changes 2 | 3 hours ago |
| .DS_Store           | Changes 2 | 3 hours ago |
| README.md           | Changes 2 | 3 hours ago |

# ***PCB Updates***

Working on placement file updates

- Noticed discrepancy with component placements and drill file
- Contacted DL-Engineering; in the process of fixing
- Will have to go to procurement office for new quote + purchasing

# Documentation Updates

## GitHub Documentation Created

- GitHub Documentation Created to maintain consistency moving forward
- Mostly in .MD files

### Monitoring

This document serves as the standard operating procedure and technical manual for deploying, operating, and managing data from the custom miniaturized Printed Circuit Board (PCB) designed to monitor keel bone dynamics in poultry.

#### 1. Hardware Specifications and Setup

##### 1.1 Physical Components

The biomedical sensor node consists of a custom four-layer PCB integrated into an animal-wearable form factor. Key components include:

- **Microcontroller:** Nordic Semiconductor nRF52840 SoC (supporting ARM Cortex-M4F and Bluetooth Low Energy 5.0 protocols).
- **Inertial Measurement Unit (IMU):** TDK InvenSense ICM-20948 9-axis MEMS sensor (3-axis gyroscope, 3-axis accelerometer, and 3-axis compass).
- **Power Supply:** Onboard rechargeable Lithium-Polymer (Li-Po) battery circuit regulated by a Microchip MCP73831 charge management controller.
- **Storage:** External SPI flash memory for high-fidelity data logging when wireless telemetry is obstructed.

##### 1.2 Encapsulation and Biocompatibility

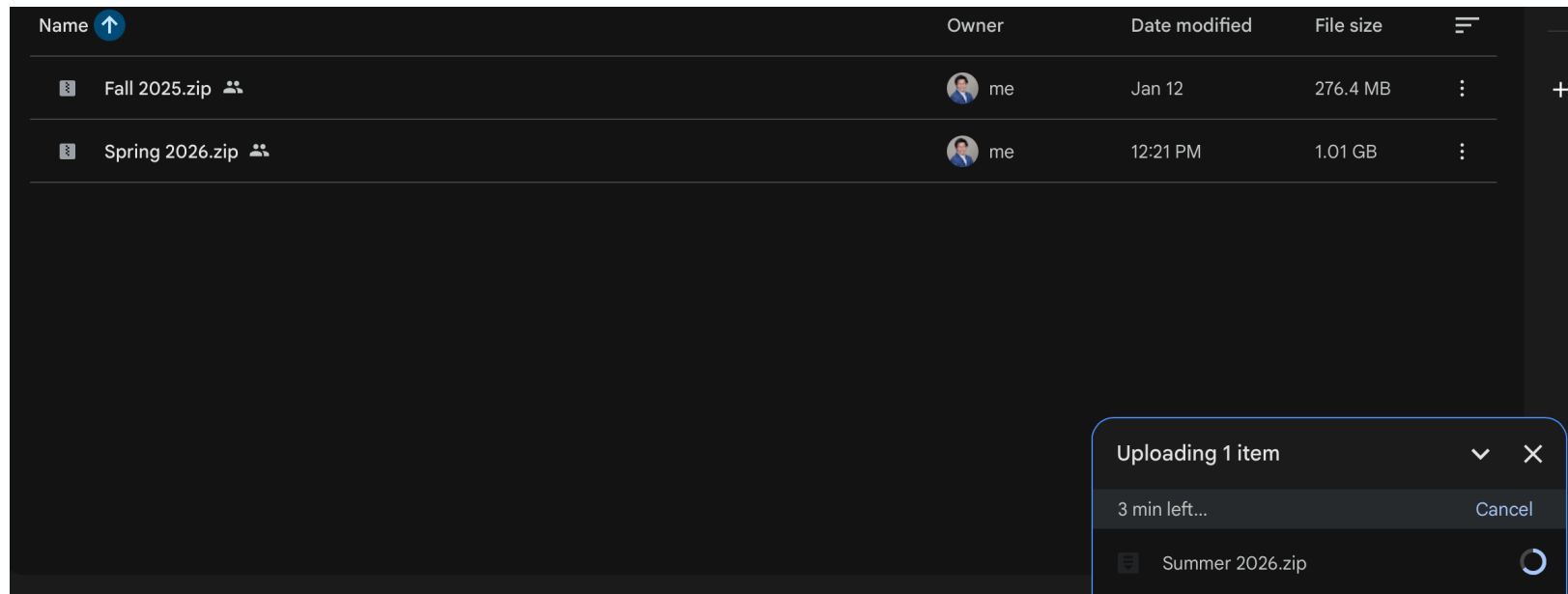
To protect the electronics from moisture, bodily fluids, and mechanical stress, the hardware must be encapsulated before deployment:

1. Ensure the battery is securely attached and the board has been programmed and post-assembly tested.
2. Verify that the total package weight does not exceed the target threshold for the specific avian subject to avoid altering natural gait.

# Documentation Updates

## GitHub Files Transferred

- Files Transferred over into Google Drive:
- [https://drive.google.com/drive/folders/1Ul3LAQrhO1t5J\\_EVqvNEP-hPAe9nVNyN](https://drive.google.com/drive/folders/1Ul3LAQrhO1t5J_EVqvNEP-hPAe9nVNyN)



# Documentation Updates

## Ownership Transferred

- Transferred ownership of repository over to Nolan
- Should take a few days

**Transfer repository: NHuang58/Biomedical\_Device\_PCB**

Transfer this repository to another user or to an organization where you have the ability to create repositories.

*Required fields are marked with an asterisk (\*).*

⚠ To understand admin access, teams, issue assignments, and redirects after a repository is transferred, see [Transferring a repository](#) in GitHub Help.

Transferring may be delayed until the new owner approves the transfer.

**New owner \***

☐ Select one of my organizations

☒ Specify an organization or username

**Type NHuang58/Biomedical\_Device\_PCB to confirm. \***

[I understand, transfer this repository.](#)

# Paper Updates

## Paper Acceptance Date Changed

- Now notification of Acceptance will be on 8/21
- No updates on author portal

**WEDNESDAY, MAY 6, 2026**  
Proposals for Workshops

**THURSDAY, JULY 2, 2026**  
Option A Paper Submission Deadline

**FRIDAY, AUGUST 21, 2026**  
Notification of Paper Acceptance

**SUNDAY, SEPTEMBER 13, 2026**  
Submission of Final Papers

### System in Egg Laying Hens

This submission is under consideration and cannot be edited. Further information will be emailed to you by the journal editorial office.

[Submission overview](#) →

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# *Next Week*

## Plans for next week

- Get PCB fully ordered and ensure payment goes through
- Wrap up documentation for next developer

# *Next Week*

## Thoughts

- Very useful interdisciplinary project where I was able to practice a lot of skills
  - Firmware Development
  - PCB Design
  - Sensor Testing
  - Communication with other parties
- Lot of potential for the project; can't wait to see where it goes

# *Thank You*

Purdue Polytechnic Institute



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